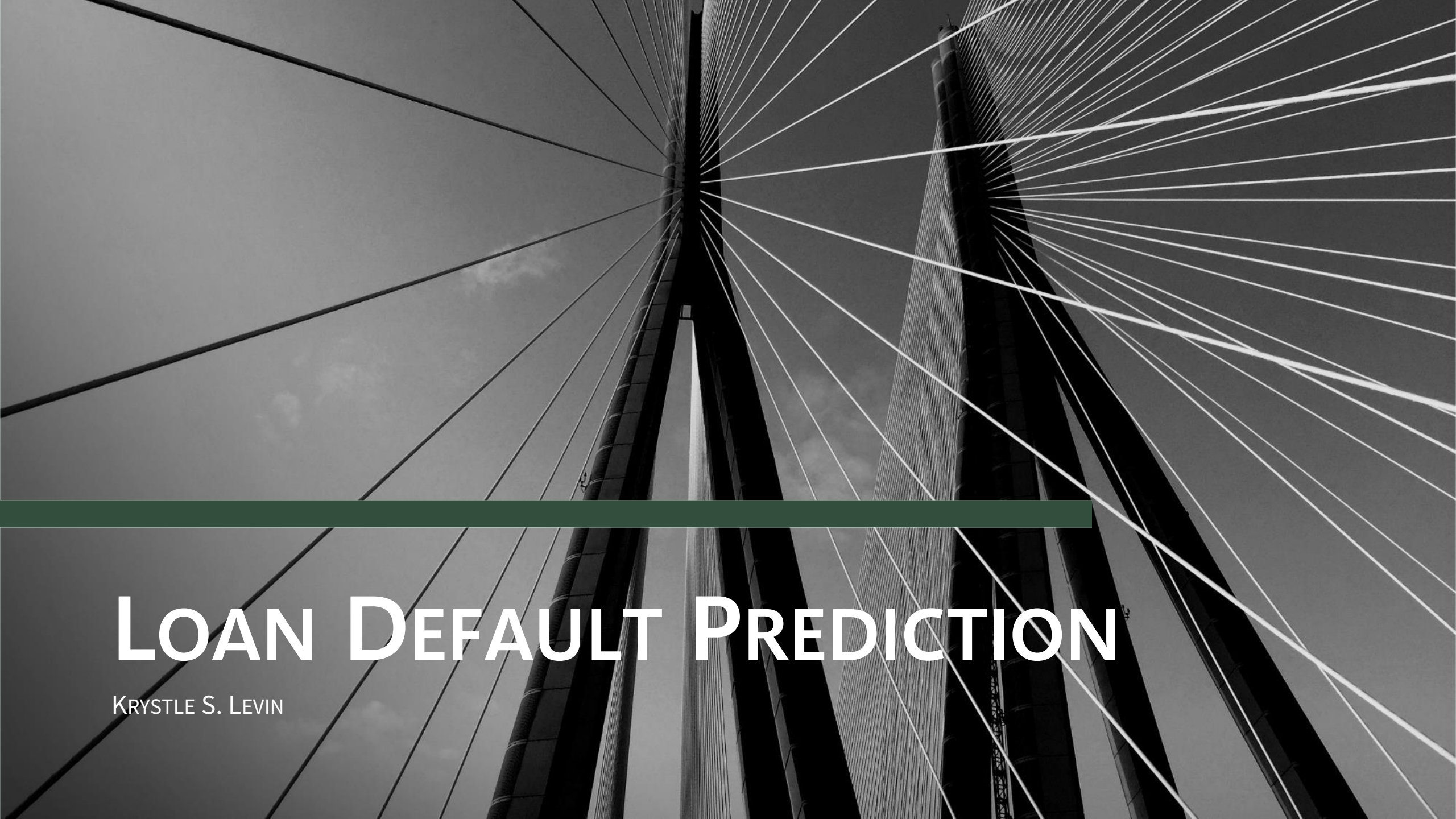




LOAN DEFAULT PREDICTION

KRYSTLE S. LEVIN

PROBLEM

LOAN APPLICATION PROCESS

Heuristic automation attempts have failed to address the multifaceted loan approval process to check applicant creditworthiness,

USABILITY

Lenders would like a model that is easy to interpret, have consistent outcomes, and expected results.

BIAS

Approximately 20% of all home equity loans result in default, posing a significant threat to lenders.

DEFAULTED LOANS

Current efforts are subject to the bias of the loan reviewers, allowing for human error and poor judgement during the approval process.

FINANCIAL LOSS

The 20% of defaulted loans equate to approximately \$20 million in potential losses



KEY BUSINESS QUESTIONS

FEATURE IMPORTANCE

What information from existing loans can we analyze to determine the dominant metrics related to loans that end in default?

EASY TO USE

What model will produce the most interpretable model for lenders to simplify the decision process and provide proof that decisions were made free of bias for any adverse behaviors or rejections?

LOSS REDUCTION

What are the applicant thresholds that lenders should adopt to decrease the number of approved loans that end in default?

PROPOSED SOLUTION:

DECISION TREE

WITH HYPER PARAMETER TUNING
AND RECALL PRIORITIZATION

INTERPRETABILITY

The main benefit of a decision tree is the interpretability of its output. A Decision Tree more closely mimics human decision making than other classification models. The open box capabilities that will allow lenders to visualize and interpret the data will help support rejections in cases of decision disputes.

TESTABILITY

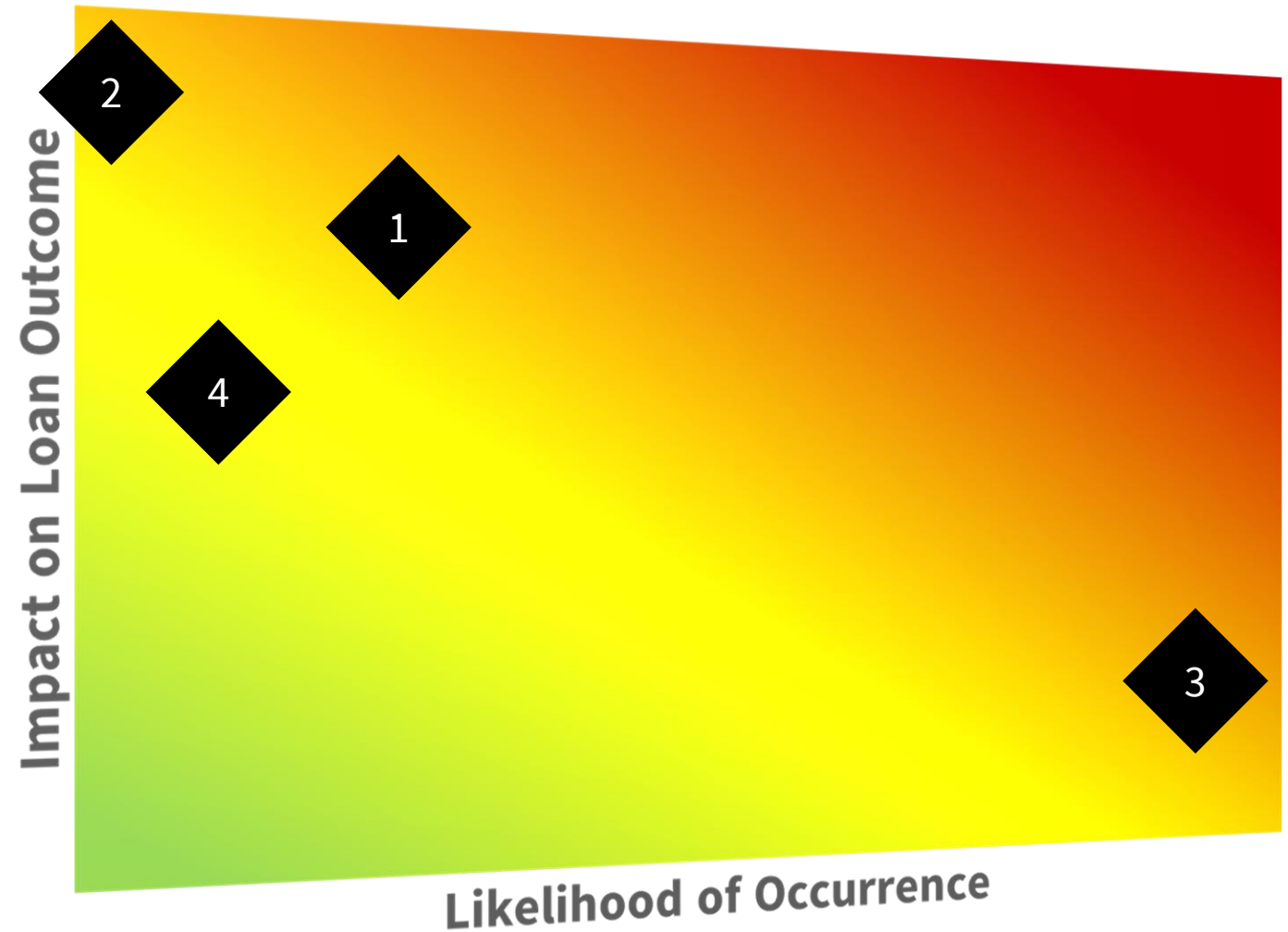
Decision Trees can easily be validated using statistical tests which can further support the justification of any disputed loan decision of rejection or otherwise.

VERSATILITY

Decision Tree can handle both numerical and categorical data while naturally deemphasizing irrelevant features. Additionally, hyperparameter pruning will allow us to optimize our model for the best results.

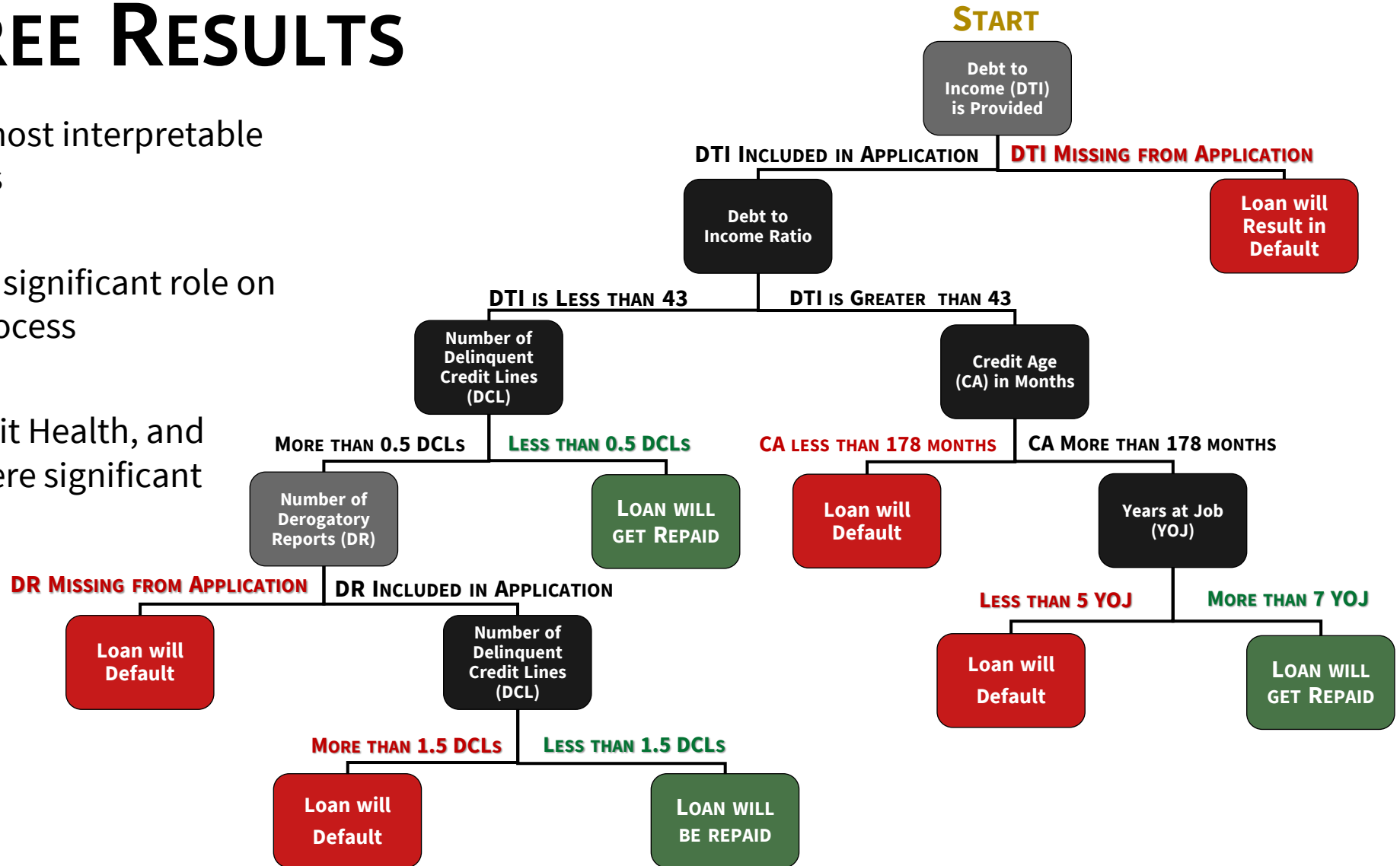
Key Insights

- 1 Borrowers with **missing debt-to-income ratios** were the most likely to result in default
- 2 Those with **higher debt to income ratios** were more at risk for default
- 3 Borrowers with a higher number of **delinquent credit lines** and high debt to income ratios were more likely to Default
- 4 Borrowers with a younger **Age of oldest credit line** and high debt to income ratios were more likely to default



DECISION TREE RESULTS

- ◆ Decision Tree had the most interpretable results of all the models
- ◆ Missing values played a significant role on the decision-making process
- ◆ Income, Debt, and Credit Health, and Employment History were significant predictive values



PROPOSED BUSINESS SOLUTION

APPLICATION REQUIREMENTS

Policies should be put in place requiring loan applications to be filled in full to be prior as a requirement for consideration

DATA IMPROVEMENTS

Capturing additional data points, such as an applicant's income and overall debt, can help provide validation for other fields, such as debt to income, to cross check against human error and bias.

SUPPORT PROGRAMS

Assistance and Educational programs can help prevent those at risk of defaulting from doing so at less cost and risk to the company.

Business Execution

LOAN APPROVALS

- ◆ Require applications to be filled in their entirety before accepting it for review
- ◆ Set debt to income ratio threshold limits as a primary criteria for acceptance
- ◆ Define and enforce policies to ensure application reviewer bias does not overwrite baseline approval thresholds
- ◆ In implementing the above there will still be risks for delinquent loans, but the amount of loss should reduce significantly

INVESTMENTS

- ◆ Investments in automation tools to make the decision process based on the applicant's details can help eliminate the bias cause by human approvals.
- ◆ Support programs for borrows, such as financial advisors, financial planning courses, and payment deferment programs can help reduce the likelihood of at risk approved borrowers from having their loan end in default.



SUMMARY

- ◆ Decision Tree models best meet the objectives of the client for producing an interpretable and simplified loan approval process
- ◆ Completeness of the application should be a requirement before a loan goes to review

- ◆ Loan applications with no listed debt to income should always be denied.
- ◆ Those with a debt to income higher than 44 and a credit age of less than 178 months should be denied if they have less than 5 years at their job. Those with an oldest credit line younger than 178 months should be denied
- ◆ If the applicant's debt to income is lower than 44, if the quantity of their derogatory reports is missing or the quantity of their delinquent credit lines exceeds 1.5, the loan should not be approved.



THANK YOU

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Appendix

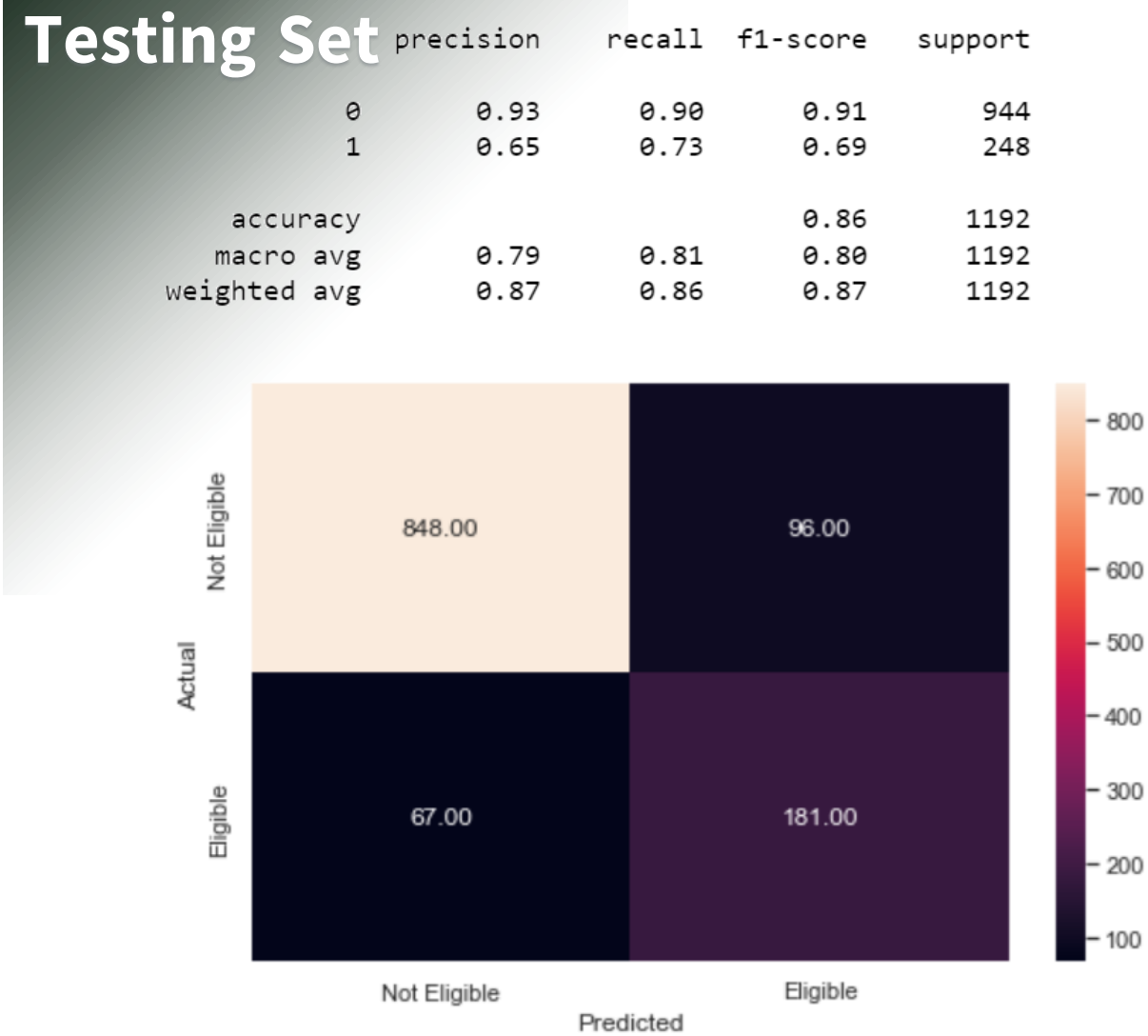
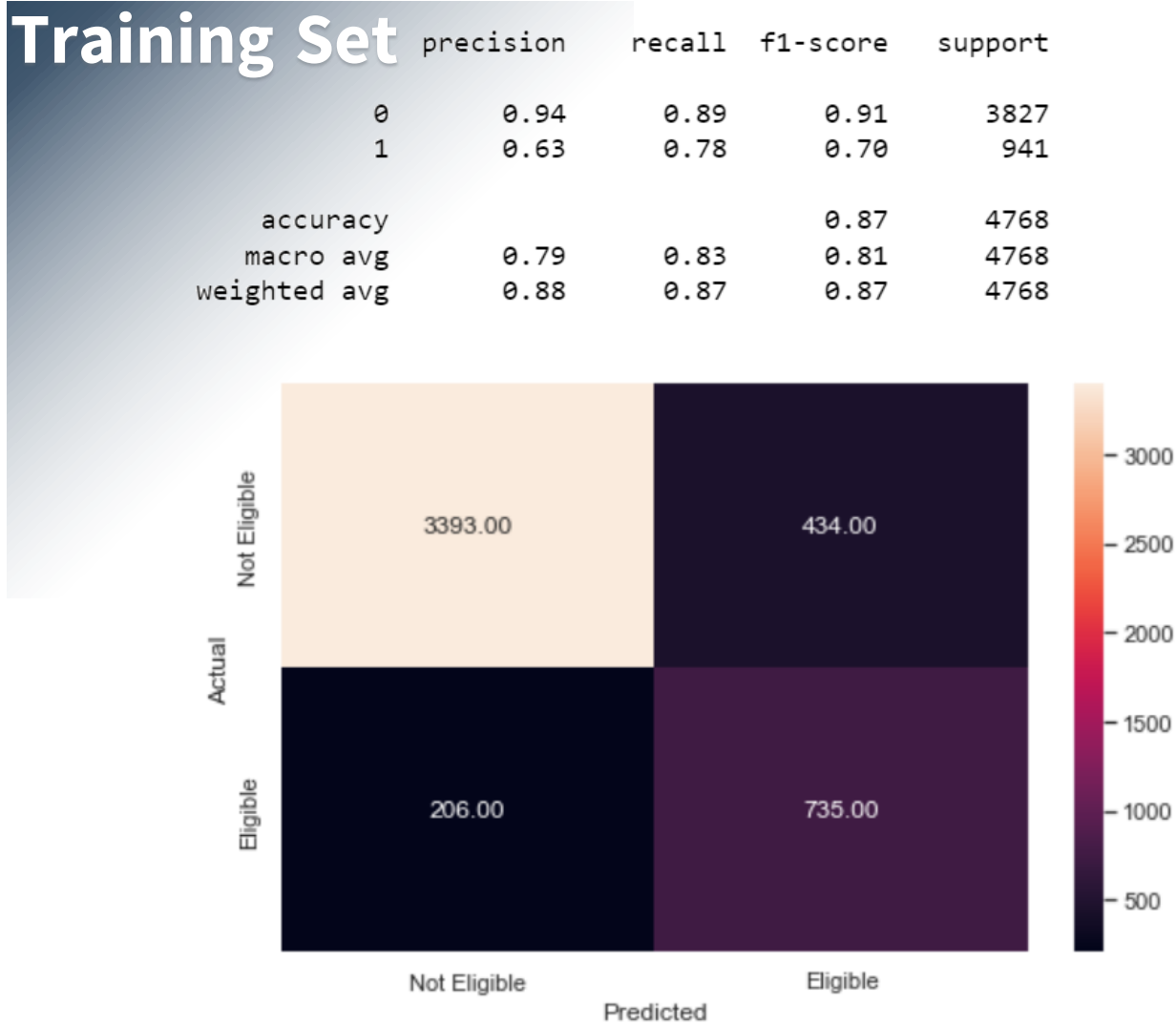
12/2022

Loan Default Prediction - Krystle S. Levin

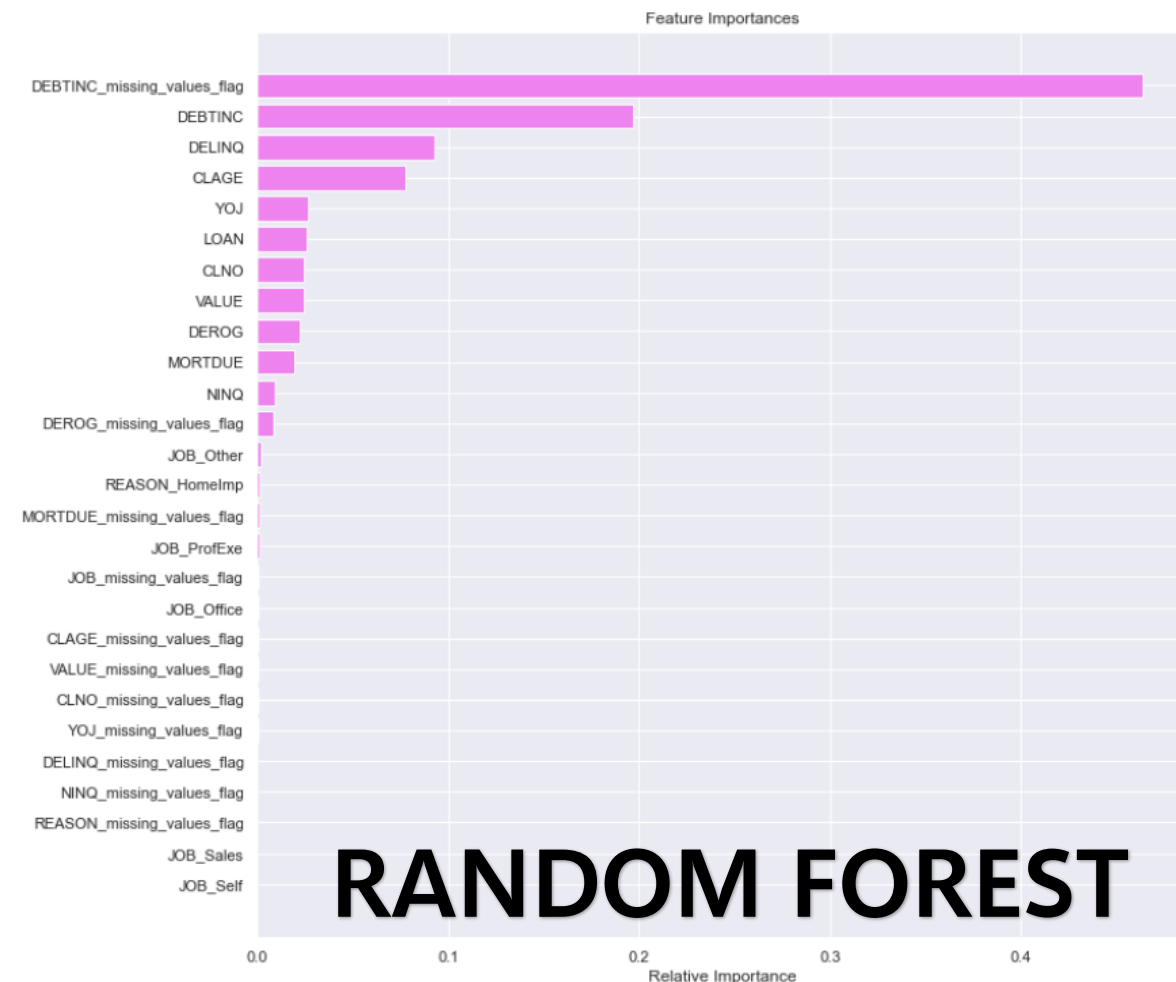
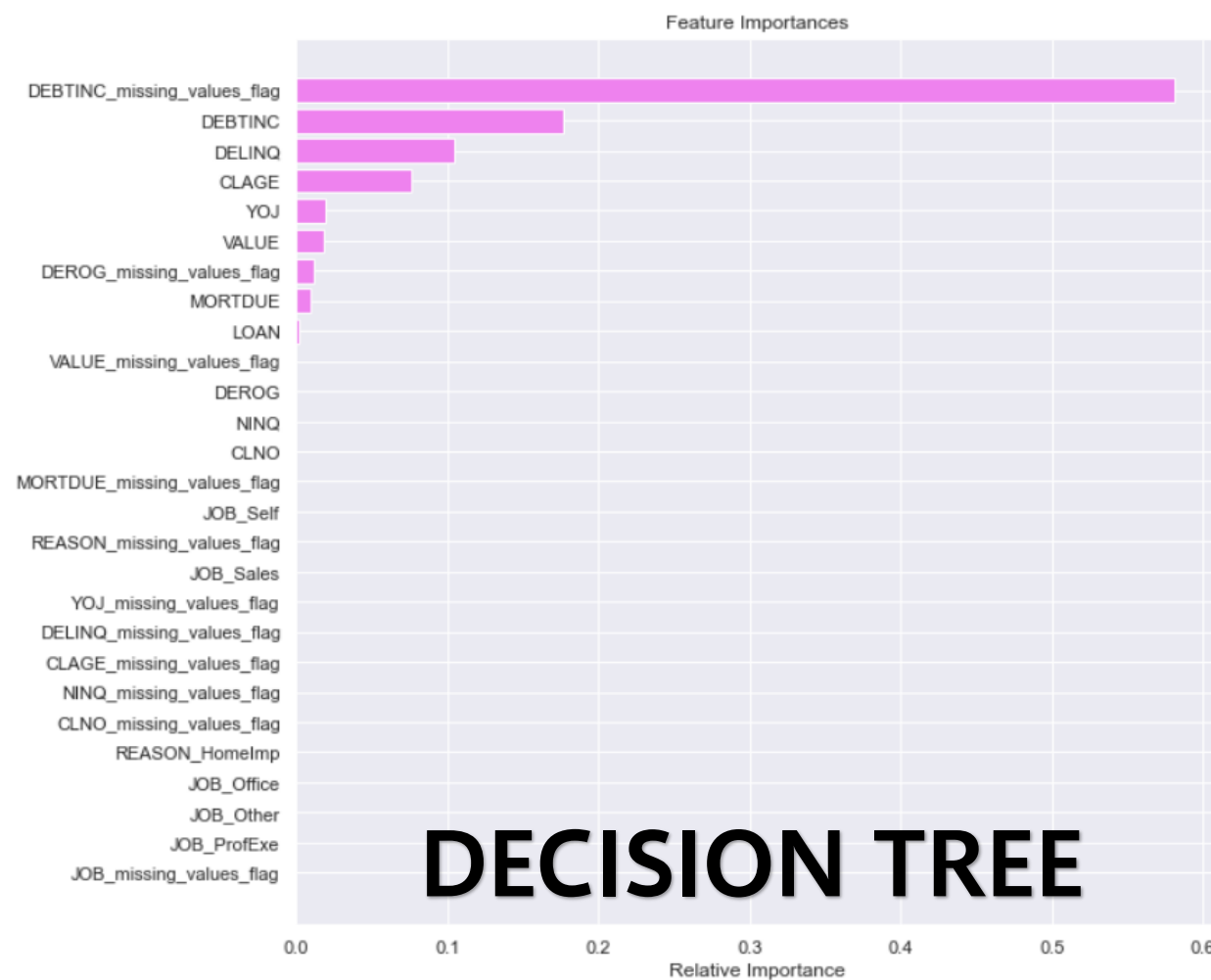
Model Comparison

		Logistic Regression	Decision Tree	Tuned Decision Tree	Random Forest	Weighted Random Forest	Tuned Random Forrestr
Train	Accuracy	0.80	1.0	0.87	1.0	1.0	0.89
	Recall	0.34	1.0	0.78	1.0	1.0	0.81
	Precision	0.71	1.0	0.63	1.0	1.0	0.69
Test	Accuracy	0.79	0.87	0.86	0.91	0.91	0.87
	Recall	0.01	0.63	0.73	0.69	0.67	0.73
	Precision	0.50	0.71	0.65	0.85	0.87	0.69

Tuned Decision Tree Performance



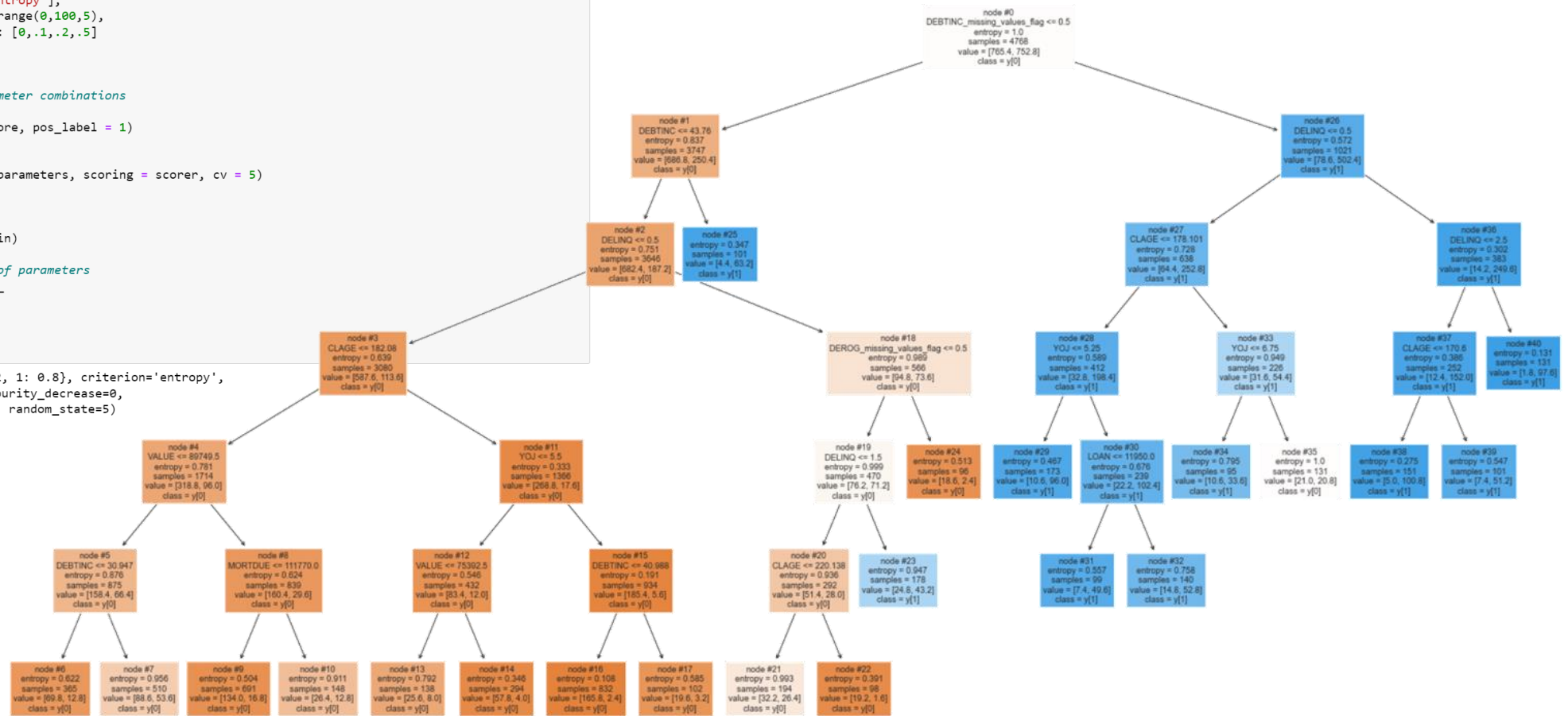
Feature Importance Comparison



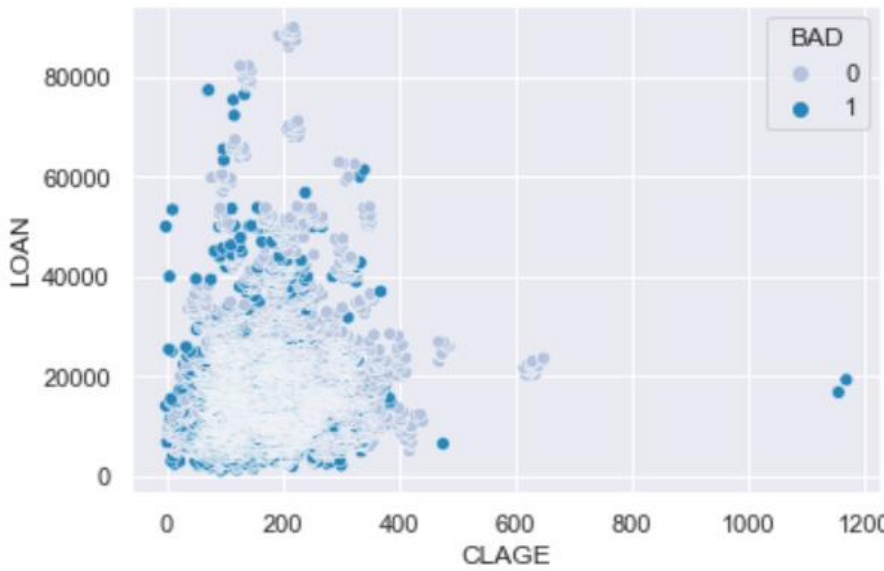
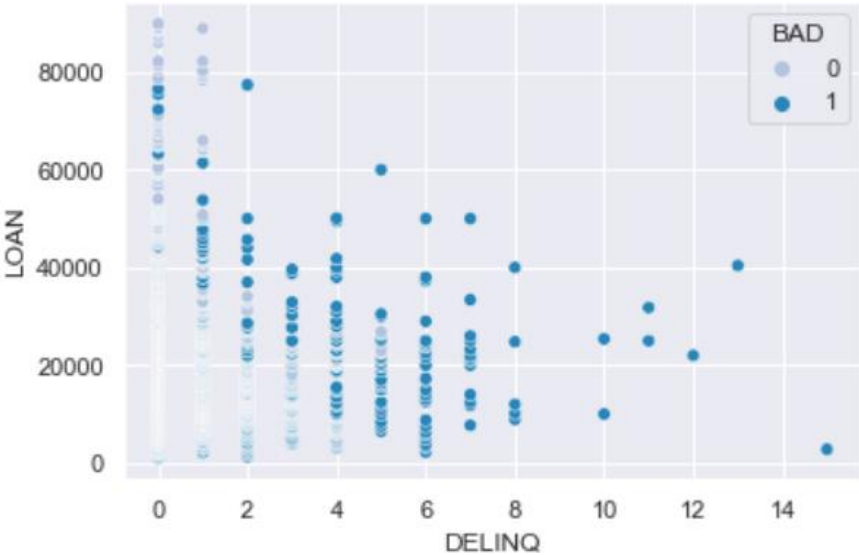
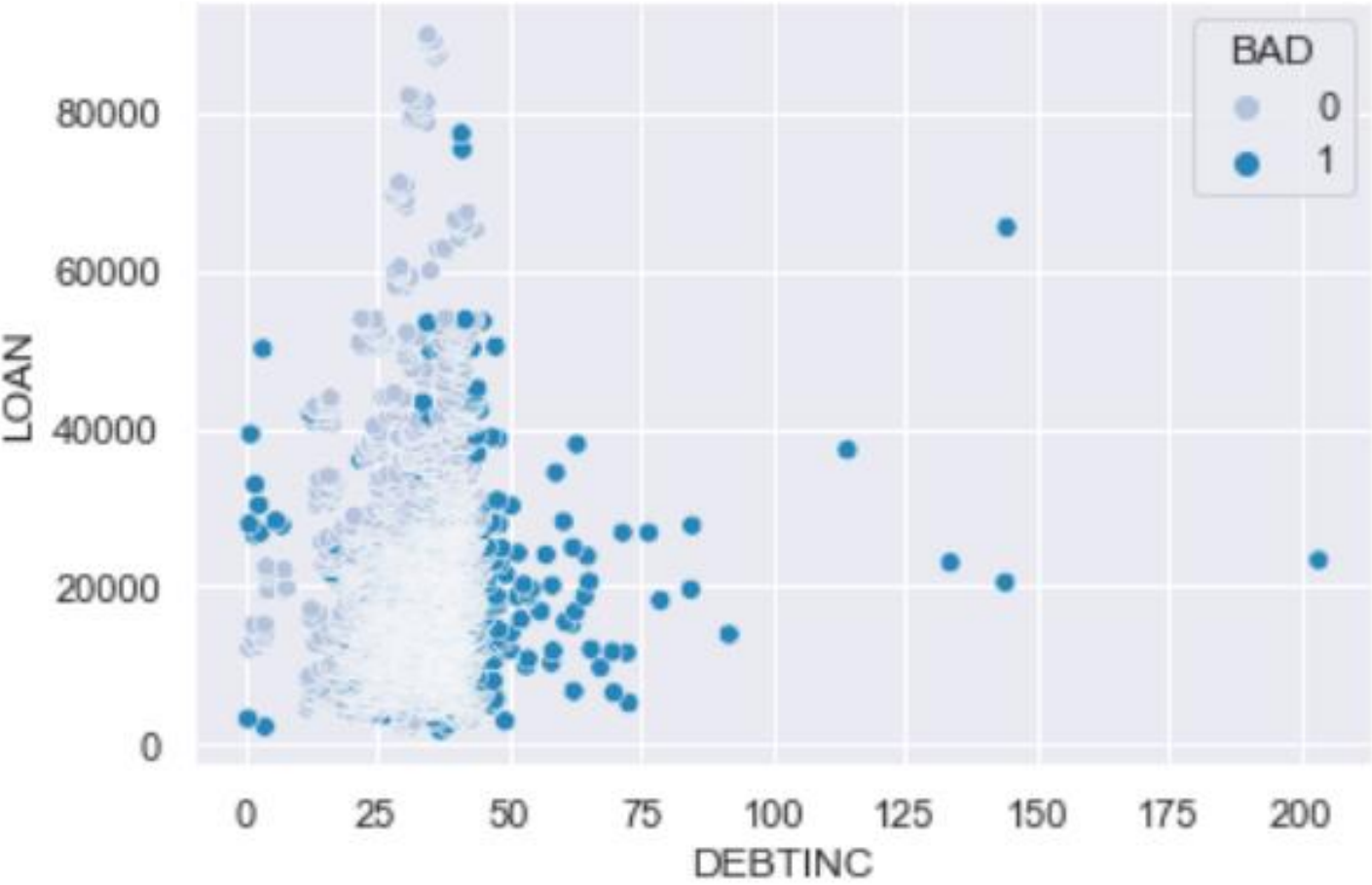
Code Highlights:

```
In [184]: 1 # Choose the type of classifier.
2
3 d_tree_tuned = DecisionTreeClassifier(random_state = 5, class_weight = {0: 0.2, 1: 0.8})
4
5
6 # Grid of parameters to choose from
7
8 parameters = {'splitter': ['best', 'random'],
9               #'random_state': np.arange(1,5),
10              'max_depth': np.arange(3,8),
11              'criterion': ['gini', 'entropy'],
12              'min_samples_leaf': np.arange(0,100,5),
13              'min_impurity_decrease': [0,.1,.2,.5]
14            }
15
16 # Type of scoring used to compare parameter combinations
17
18 scorer = metrics.make_scorer(recall_score, pos_label = 1)
19
20 # Run the grid search
21 grid_obj = GridSearchCV(d_tree_tuned, parameters, scoring = scorer, cv = 5)
22
23
24 # Fit the GridSearch on train dataset
25 grid_obj = grid_obj.fit(X_train, y_train)
26
27 # Set the clf to the best combination of parameters
28 d_tree_tuned = grid_obj.best_estimator_
29
30 # Fit the best algorithm to the data.
31
32 d_tree_tuned.fit(X_train, y_train)
```

```
Out[184]: DecisionTreeClassifier(class_weight={0: 0.2, 1: 0.8}, criterion='entropy',
max_depth=6, min_impurity_decrease=0,
min_samples_leaf=95, random_state=5)
```



MULTIVARIANT ANALYSIS OF LOAN OUTCOMES & AMOUNTS BY IMPORTANT FEATURES



Probabilities of 4 Most Important Features

Feature Metric	# of Occurrences	P(Default, Given the Feature has Occurred)
DEBTINC is missing	1267	60%
DEBTINC is greater than 44	103	85%
CLAGE is less than 288 Months	5037	21%
DELINQ is greater than 1.5	547	57%

Two Feature Metrics	# of Occurrences	P(Default, Given Both Features Occurred)
DEBTINC > 44 and CLAGE < 288	96	89%
DEBTINC > 44 and CLAGE < 288	19	100%